MIL-D-80002D
21 October 1982
SUPERSEDING
MIL-D-80002C
31 August 1976

MILITARY SPECIFICATION

DRILLING MACHINE, UPRIGHT FLOOR TYPE, ROUND AND BOX COLUMN, SLIDING HEAD, PLAIN TABLE, POWER FEED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven, floor mounted, round and box column, sliding head, power feed, single and multiple spindle, upright drilling machines with equipment as described herein.

1.2 Classification. This specification covers drilling machines of the following types, styles, and sizes. The type style, and size required for specific procurement shall be as specified (see 6.2.1). All sizes listed below are available in Type I and Type II machines.

Type I - Round or Box Upper Column, Round Lower Column
  Style A - One spindle

Type II - Round or Box Upper Column, Box Lower Column
  Style A - One spindle
  Style B - Two spindles
  Style C - Three spindles
  Style D - Four spindles
  Style E - Five spindles
  Style F - Six spindles

Size 22 - 1-1/4 (31.75mm) inch drill capacity - 22 (50.80mm) inch swing
Size 24 - 2 (63.50mm) inch drill capacity - 24 (76.20mm) inch swing
Size 26 - 2-1/2 (558.80mm) inch drill capacity - 26 (609.60mm) inch swing
Size 28 - 3 (660.40mm) inch drill capacity - 28 (711.20mm) inch swing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Propsal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3413

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitations form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment

MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

2.1.2 Other Government documents, drawings and publications. The following Government documents, drawings and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards 29 CFR 1910

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B5.1 - T-Slots, Their Bolts, Nuts, Tongues, and Cutters
B5.10 - Machine Tapers, Self-Holding and Strip Taper Series
B11.8 - Drilling, Milling and Boring Machine Construction Care and Use of Safety Requirements

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C113.1 - Electrical Standards for Metalworking Machine Tools (NFPA No. 79)
Z210.1 - Metric Practices (ASTM-E-380)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY, 10018.)

National Electrical Manufacturers' Association Standards (NEMA)

NEMA MC-1 - Motors and Generators

NEMA IC-1 - Industrial Controls

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th Street, New York, NY 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample unit for first article inspection and approval (see 6.2.1 and 6.3).

3.2 Design. The drilling machines shall be new and one of the manufacturer's current models capable of drilling work pieces within its specified capacity in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the U. S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ANSI Z210.1.

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3.2.1.1 Measuring and indicating device calibrations. When specified (see 6.2.1), measuring and indicating devices such as feed screw dials, scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and all other similar devices shall be graduated in the specified system (US or SI) of measurements. Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and SI System of Measurements. When a dual US and SI system is furnished, both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The drilling machine shall contain reclaimed materials to the maximum extent possible without jeopardizing its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition and physical properties as the materials originally selected for use on the machine. Reclaimed materials shall include iron, steel, copper, brass, aluminum, fiber products, plastics, and elastomers that have been collected from discarded solid, liquid, semi-solid, or gaseous waste such as garbage, refuse, or sludge.

3.2.3 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. In addition, the machine shall comply with all requirements of ANSI B11.8 that are designated therein as the responsibility of the machine manufacturer. In the event of conflict between the requirements of the OSHA and ANSI Standards, the requirements of OSHA shall apply. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.5 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Each recirculating system shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

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3.2.6 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The drilling machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Casting and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts and other fasteners shall be installed in a manner that prevents change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the drilling machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than 0.001 inch. Dial diameters shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.3.8 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

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3.3.9 Gears. All gears shall be machined from a material suitable for the intended purpose. All gears shall be heat treated by a process that will impart the hardness and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature wear.

3.4 Components.

3.4.1 Coolant system. Unless otherwise specified (see 6.2.1), a coolant system shall be provided. The coolant system shall include a sump or reservoir, a power driven pump, and piping. The coolant reservoir shall be of sufficient capacity to permit full flow cooling. The coolant system shall include a baffle(s) or strainer which is easily removable for cleaning. The reservoir shall be equipped with means for emptying and cleaning. A coolant delivery nozzle and valve shall be provided to permit the operator to direct and control the amount of coolant flow. Means for returning coolant from work area to the reservoir shall be provided.

3.4.2 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to all requirements in this paragraph. The system shall conform to the requirements of ANSI/NFPA-79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit having a fused safety disconnect switch. The drilling machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in Table 240-A of ANSI/NFPA-79 for the disconnect fuse rating.

3.4.2.1 Motors. Motors shall have ball or roller bearings. Continuous duty horsepower ratings shall conform to the requirements of Table I. Unless other specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure.

3.4.3 Base. The base shall be an iron or iron alloy casting, or a steel weldment, ribbed and braced to provide a foundation for the entire machine. Means shall be provided for leveling and securing the machine to a flat mounting surface. The coolant fluid sump or reservoir, as required under 3.4.1, shall be included as an integral part of the base or as a separate external unit. That portion of the base that is below the spindle/quill assembly on the Type I machine, shall also serve as an auxiliary working surface for accommodating extra large workpiece sizes. The auxiliary working surface shall be machined to not rougher than 125 microinch finish and have at least two parallel T-slots. Machined T-slots shall be in accordance with ANSI B5.1. The periphery of the auxiliary working surface shall have a coolant fluid trough.

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3.4.4 Column. The column shall be an iron or iron alloy casting mounted on the base and reinforced to provide support and minimize deflection. Machines with cylindrical columns shall be cross sectionally round on the lower column with the exterior surface machined for vertical adjustment of the worktable; and rectangular upper columns with machined guideways for vetical adjustment of the sliding head. Machines with box columns shall have machined guideways for vertical adjustment of the worktable and sliding head.

3.4.5 Worktable. The worktable shall be an iron or iron alloy casting, ribbed and reinforced to withstand maximum capacity workpiece weights and tool thrust loads. The working surface of the worktable shall be machined to a finish of not rougher than 125 microinches, level, and of the size required herein. Worktable height adjustment shall be accomplished by a mechanical drive arrangement of either a crank and worm gear, crank and screw-jack or crank and rack and pinion type mechanism. The working surface of the worktable for the Type I machine shall be round as viewed from the top, rotatable around its center, and revolvable around the column to a position necessary to clear the auxiliary working surface of the base. The working surface of the worktable for the Type II machine shall be rectangular with a coolant fluid trough around its periphery with means for returning coolant to the reservoir. Means shall be provided for securing the worktable in any position within its adjustment, rotating or revolving ranges. The working surface of the worktable shall have the number and arrangement of T-slots normally furnished by the manufacturer. Machined T-slots shall be in accordance with ANSI B5.1.

3.4.6 Top head and spindle drive. Unless otherwise specified (see 6.2.1), the top head shall house the spindle drive assembly in a stationary position at the top of the column. The spindle shall be driven by an assembly of motor-driven mechanical components with shafts, gears, belts, or pulleys as necessary for developing and maintaining spindle rotating speeds and feed rates as required herein. All rotating and sliding components shall be mounted on bearings or bushings with means for positive and continuous lubrication.

* 3.4.7 Sliding head. The sliding head shall be an iron or iron alloy casting or a steel weldment, arrangement to support the spindle/quill assembly. The head shall be mounted and vertically adjustable along the guideways of the upper column. Means shall be provided for adjustment to take up guideway wear. Clamps shall be provided for locking the sliding head in position after adjustment. The sliding head shall be counter-balanced to facilitate ease of adjustment.

* 3.4.7.1 Spindle/quill assembly. The quill shall be machined of steel alloy, and shall be fitted to the sliding head. Vertical travel of the quill shall be accomplished by means of a rack and pinion drive. Quill position shall be indicated by either a graduated scale engraved on the

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quill in increments of not more than 1/8 inch, or by a feed depth dial. The spindle shall be machined from steel alloy and shall rotate within the quill on ball or roller bearings. The characteristics of the spindle shall comply with ANSI B5.10. The quill/spindle assembly shall be counterbalanced and be power fed for accomplishing performance specified herein. The spindle feed shall be obtained through a variable speed motor drive or a gear drive arrangement which is hand lever operated through a positive engagement type clutch. An adjustable spindle depth trip shall be provided to disengage power feed at preset depth. The spindle shall be equipped with an automatic safety trip for power feed at bottom limit of spindle travel.

3.4.8 Worklight. The machine shall have a worklight with a lamp and protective shield. The worklight design shall permit it to be attached to the machine and be swung over the tool and workpiece. The worklight shall have an on-off switch.

3.4.9 Multiple spindle machines. Multiple spindle machines shall comprise a plurality of single spindle machines, as specified herein, and be mounted on a common base. Each spindle shall operate as an independent unit and each shall be interchangeable with each of the other units. Unless otherwise specified, the spindle center distance and the table size shall be in accordance with Table I.

3.5 Standard equipment. Unless otherwise specified (see 6.2.1), one set of standard equipment or tools normally provided by the manufacturer for making machine adjustment shall be furnished.

3.6 Optional equipment. Optional equipment and accessories shall be furnished as specified (see 6.2.1).

3.7 Sizes and capacities. The sizes and capacities of the drilling machine shall conform to the requirements of Table I.

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TABLE I

Sizes and Capacities

CHARACTERISTIC	REQUIREMENTS							
	TYPE I				TYPE II			
	22	24	26	28	22	24	26	28
Drilling capacity, cast iron, inches	1-1/4	2	2-1/2	3	1-1/4	2	2-1/2	3
Distance of face of found column to spindle center, inches	11	12	13	14	-	-	-	-
Distance box-column ways to spindle center, inches	-	-	-	-	11	12	13	13-1/2
Swing, drills to center of circle, diameter, inches	22	24	26	28	22	24	26	
Distance from spindle nose to table, inches	23	23	24	25	23	23	24	25
Distance from spindle nose to finished surface of base inches	42	42	42	46	-	-	-	-
Vertical travel of spindle/quill assembly, inches	5	5	8	8	5	5	8	8
Working surface of base, width x length, inches	16 x 16	20 x 22	20 x 22	23 x 24	-	-	-	-
Number of feeds	4	4	6	6	4	4	6	6
Rates, spindle feed, inches per revolution	.004 to .016	.006 to .018	.006 to .018	.005 to .016	.004 to .016	.006 to .018	.006 to .018	.005 to .016
Number of spindle speeds	4	6	8	8	5	6	8	8
Spindle speed range, RPM	65 to 1350	110 to 1100	80 to 1000	80 to 1000	65 to 1350	100 to 1100	80 to 1000	80 to 1000
Diameter, round column, inches	5-1/2	6-1/2	8	8	-	-	-	-
Spindle nose, diameter, inches	1-5/8	2	2-1/2	2-1/2	1-5/8	2	2-1/2	2-1/2
Spindle bored to fit morse taper	No. 4	No. 4	No. 5	No. 4	No. 4	No. 4	No. 5	No. 4
Diameter quill, inches	2-1/2	2-9/16	2-3/4	3-1/2	2-1/2	2-9/16	2-3/4	3-1/2
Table diameter, inches	17	23	26	27	-	-	-	-
Table working surface, width x length, inches								
Style A	-	-	-	-	22 x 22	19 x 24	20 x 22	21 x 24
Style B	-	-	-	-	22 x 38	19 x 36	20 x 36	21 x 36
Style C	-	-	-	-	22 x 54	19 x 56	20 x 58	21 x 60
Style D	-	-	-	-	22 x 70	19 x 72	20 x 76	21 x 78
Style E	-	-	-	-	22 x 86	19 x 88	20 x 90	21 x 92
Style F	-	-	-	-	22 x 102	19 x 104	20 x 110	21 x 112
Distance between spindle centers inches					11	14	16	16
Motor, horsepower	3	5	7-1/2	7-1/2	3	5	7-1/2	7 1/2

Requirements are not less than those shown. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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3.8 Performance. The machine shall be capable of drilling holes through metal solids to include cast iron for accomplishing maximum capacity hole sizes, and spindle speed and feed rate required by Table I.

3.9 Alignment and accuracy tolerances. Each machine shall meet the alignment and accuracy tolerances as specified in Table II.

TABLE II

Alignment and Accuracy Tolerances

CHARACTERISTIC	Maximum Allowable Deviation
Working surface of base, flat (Type I, Style A)	.0016"/12"
Working surface of worktable, flat	.0016"/12"
Column guideways square with working surface of the worktable, in longitudinal and tranverse planes	.003"/12"
Spindle taper, runout 1 inch from nose	.001" TIR
Spindle taper, runout 6 inches from nose	.002" TIR
Spindle square with working surface of the worktable, turn around method with 6-inch radius, in longitudnal and transverse planes	.003"/12"

3.10 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The chart or plate shall contain the following information:

Points of lubricant application
Service interval
Type of lubricant
Viscosity

3.11 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate shall be securely attached to each drilling machine. The nameplate shall contain the information listed below. If the

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drilling machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture
U S

3.12 Electromagnetic interface control. Unless otherwise specified (see 6.2.1), equipment procured under this specification shall comply with MIL-STD-461.

3.13 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.14 Workmanship. Workmanship of the drilling machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the first article.

4.3 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the test in 4.5 through 4.5.3, and the inspection in 4.6. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

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4.4 Examination. The drilling machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.9 and 3.11 through 3.13.

4.5 Tests.

4.5.1 Operational test. The drilling machine shall be operated at no load for not less than 15 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.5.2 Alignment accuracy tests. Each machine shall be tested to determine that it complies with the alignment accuracy requirements of Table II.

4.5.3 Performance test. Each machine shall be tested by drilling three holes through a solid, one-inch thick piece of cast iron as follows:

Machine	Drill Size	Spindle Speed (not less than)	Spindle Feed (not less than)
Size 22	1-1/4 inch	270 RPM	0.009 In/Rev
Size 24	2 inch	150 RPM	0.015 In/Rev
Size 26	2-1/2 inch	120 RPM	0.015 In/Rev
Size 28	3 inch	100 RPM	0.015 In/Rev

Failure of the machine to maintain the above specified minimum spindle speed and feed rate during the performance test, or any permanent distortion of machine components resulting from the test, shall be cause for rejection.

4.5.4 Electromagnetic interference control test. Unless otherwise specified (see 6.2.1), equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.12 in accordance with Table II of MIL-STD-461.

4.6 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), preservation, packing and marking shall conform to the requirements of MIL-STD-1188.

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6. NOTES

6.1 Intended use. The drilling machines covered by this specification are intended for use in production shops where wide varieties of drilling operations are performed.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement indentified to each of the following:

a. Title, number, and date of this specification.

b. Type, class, size, and style required (see 1.2).

c. When a first article is required for inspection and approval (see 3.1, 4.2 and 6.3).

d. If machine is required to be configured in specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI) state required system (see 3.2.1.1).

f. Dual calibrations (US and SI); if required (see 3.2.1.2).

g. Specify additional safety and health requirements (see 3.2.4).

h. Painting, if different (see 3.3.5).

i. Electrical system, if different (see 3.4.2).

j. Type motor enclosure, if different (see 3.4.2.1).

k. Type spindle drive, if different (see 3.4.6)

l. Standard equipment, if different (see 3.5).

m. Optional equipment, if required; fully describe required optional equipment (see 3.6).

n. Lubrication chart or plate, if different (see 3.10).

o. Nameplate, if different (see 3.11).

p. Electromagnetic interference control, if required (see 3.12)

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q. First article inspection, if different (see 4.2)

r. Quality conformance inspection, if different (see 4.3)

s. Electromagnetic interference control test requirements, if different (see 4.5.4)

t. Preservation, packing and marking, if different (see 5.1)

6.2.2 Contract data requirements. Required technical data, such as operators' manuals, part lists, wiring diagrams, and other instructions for operation and maintenance as identified on a numbered DD Form 1664 should be specified on a DD Form 1423 incorporated in the contract.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.4 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the procuring agency is advised to analyze the existing hazards levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. If specific point-of-operation guarding is required, the procuring agency should specify the exact configuration of the guard required as in most cases the guard configuration is dependent on the size and configuration of the workpieces that will be machined. The above, and any other additional safety health requirements should be specified in detail under 6.2.1(g).

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6.5 Styles and sizes in relation to previous issue. Styles in this specification are identical with the previous isssue with the exception of machines with swing sizes of 20, 21 and 23. These sizes are dropped because of lack of manufacturers of these sizes.

Custodians:
ARMY AL
Navy-SH
Air Force-99

Review activities:

Navy-SH
Air Force-84
DLA-GS

User activities:

Army-SM,
Navy-YD, MC
Air Force-84

Preparing activity:
DLA-IP

Project Number
3413-0126

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-D-80002D; Drilling Machine, Upright Floor Type, Round and Box Column, Sliding Head, Plain Table, Power Feed

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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